Industrial Internship Report on

" CREDIT CARD FRAUD DETECTION AI ENGINE"

Prepared by

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Executive Summary

This report provides details of the Industrial Internship provided by upskill Campus and The IoT Academy in collaboration with Industrial Partner UniConverge Technologies Pvt Ltd (UCT).

This internship was focused on a project/problem statement provided by UCT. We had to finish the project including the report in 6 weeks' time.

My project was "Credit Card Fraud Detection AI Engine".

This internship gave me a very good opportunity to get exposure to Industrial problems and design/implement solution for that. It was an overall great experience to have this internship.

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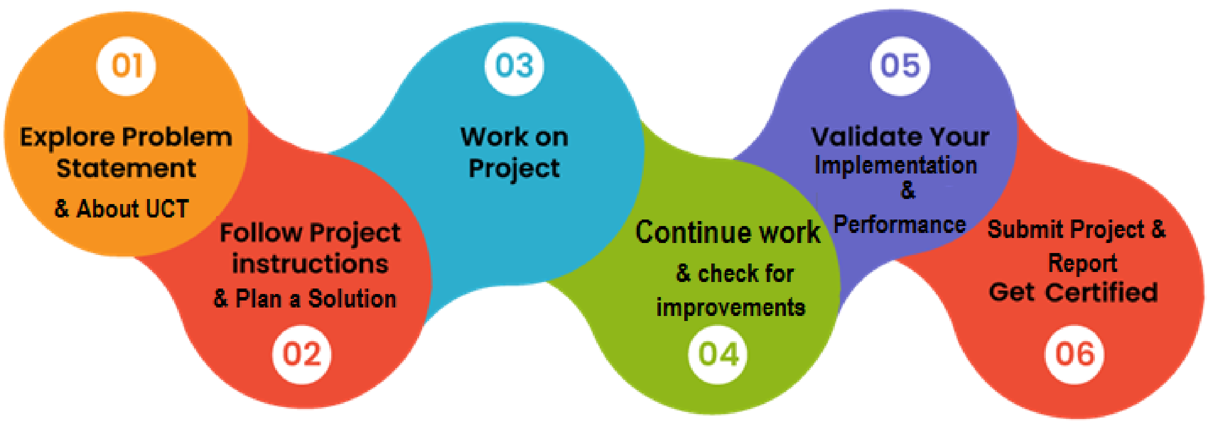
1 Preface

This report summarizes my six-week journey during the **Free Summer Internship in Data Science and Machine Learning** conducted by **upSkill Campus** in collaboration with **UniConverge Technologies Pvt. Ltd. (UCT)**. The internship provided valuable industry exposure and enhanced my technical skills in data science, machine learning, and software development.

My project, **Credit Card Fraud Detection AI Engine**, focuses on detecting fraudulent credit card transactions using machine learning. The system analyzes transaction details and predicts fraudulent activities through a Flask-based web application.

Throughout the internship, I improved my knowledge of **Python, Flask, data preprocessing, feature scaling, machine learning algorithms, and model evaluation**. I also developed stronger analytical and problem-solving skills.



I sincerely thank **upSkill Campus, UCT, my mentors, and my faculty members** for their continuous guidance and support. This internship has strengthened my confidence and prepared me for future opportunities in **Data Science and Machine Learning**. I encourage my juniors to participate in such internships to gain practical knowledge and industry experience.

2 Introduction

2.1 About UniConverge Technologies Pvt Ltd

A company established in 2013 and working in Digital Transformation domain and providing Industrial solutions with prime focus on sustainability and RoI.

For developing its products and solutions it is leveraging various **Cutting Edge Technologies e.g. Internet of Things (IoT), Cyber Security, Cloud computing (AWS, Azure), Machine Learning, Communication Technologies (4G/5G/LoRaWAN), Java Full Stack, Python, Front end** etc.

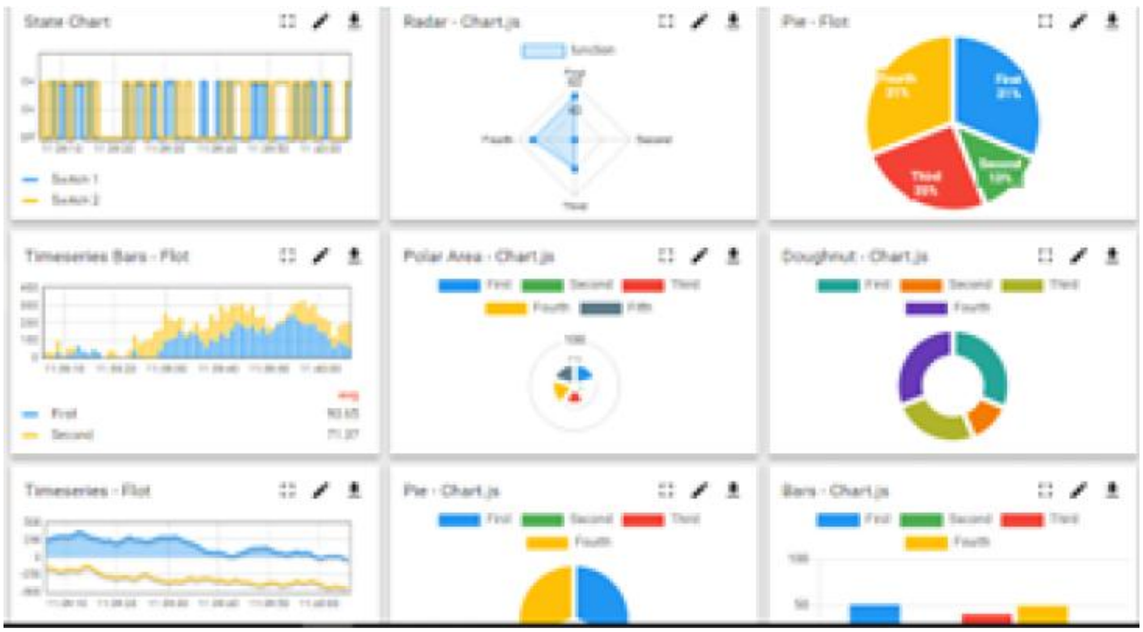
i. UCT IoT Platform ( )

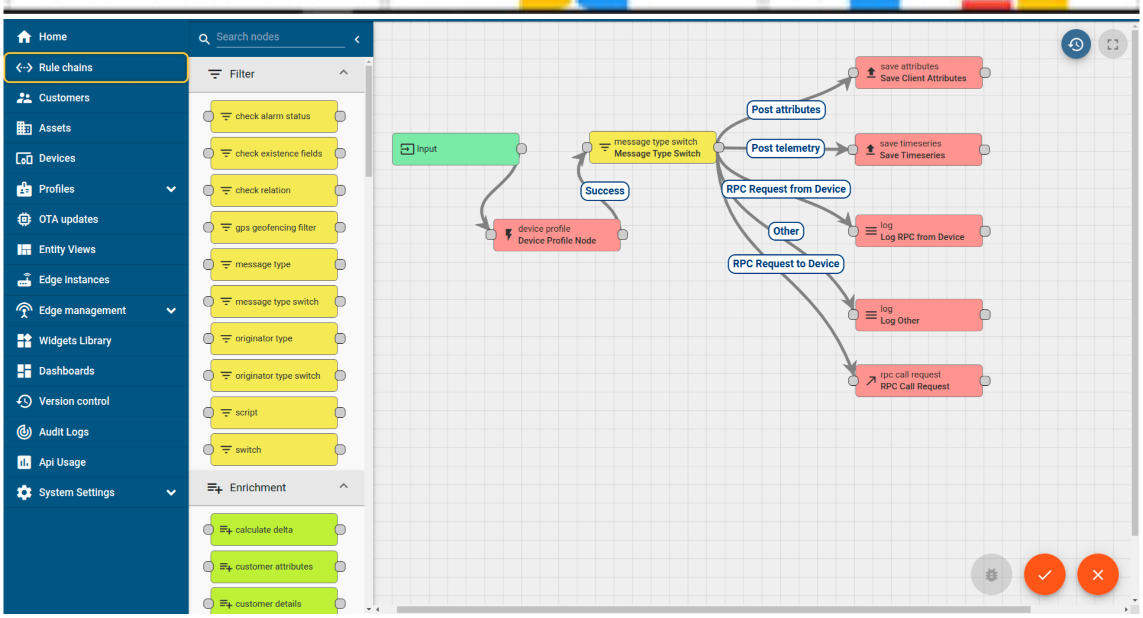
UCT Insight is an IOT platform designed for quick deployment of IOT applications on the same time providing valuable "insight" for your process/business. It has been built in Java for backend and ReactJS for Front end. It has support for MySQL and various NoSql Databases.

- It enables device connectivity via industry standard IoT protocols - MQTT, CoAP, HTTP, Modbus TCP, OPC UA
- It supports both cloud and on-premises deployments.

It has features to

• Build Your own dashboard
• Analytics and Reporting
• Alert and Notification
• Integration with third party application(Power BI, SAP, ERP)
• Rule Engine





ii. Smart Factory Platform (FACTORY WATCH)

Factory watch is a platform for smart factory needs.

It provides Users/ Factory

- with a scalable solution for their Production and asset monitoring
- OEE and predictive maintenance solution scaling up to digital twin for your assets.
- to unleased the true potential of the data that their machines are generating and helps to identify the KPIs and also improve them.
- A modular architecture that allows users to choose the service that they what to start and then can scale to more complex solutions as per their demands.

Its unique SaaS model helps users to save time, cost and money.

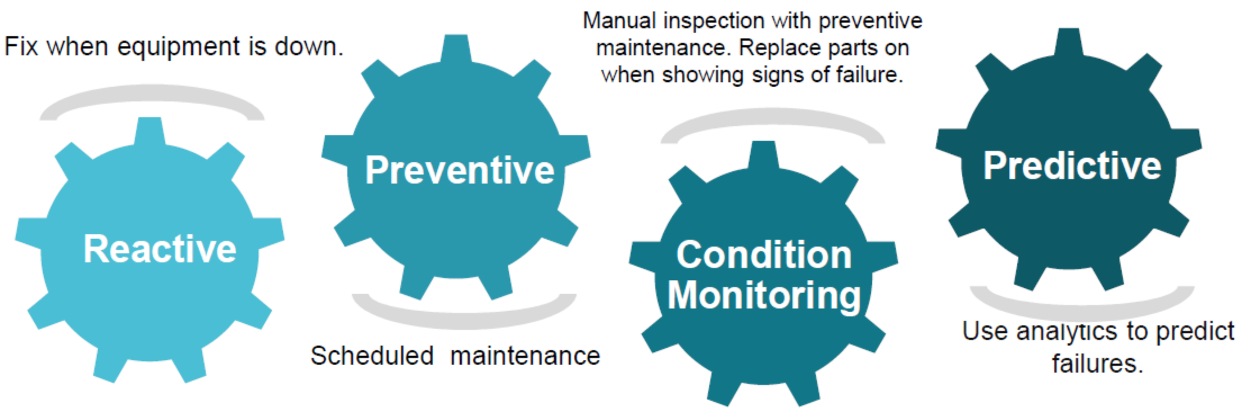
Machine	Operator	Work Order ID	Job ID	Job Performance	Job Progress		Output		Rejection	Time (mins)				Job Status	End Customer
					Start Time	End Time	Planned	Actual		Setup	Pred	Downtime	Idle		
CNC_S7_81	Operator 1	WO0405200001	4168	58%	10:30 AM		55	41	0	80	215	0	45	In Progress	i
CNC_S7_81	Operator 1	WO0405200001	4168	58%	10:30 AM		55	41	0	80	215	0	45	In Progress	i

iii. LoRaWAN based Solution

UCT is one of the early adopters of LoRAWAN teschnology and providing solution in Agritech, Smart cities, Industrial Monitoring, Smart Street Light, Smart Water/ Gas/ Electricity metering solutions etc.

iv. Predictive Maintenance

UCT is providing Industrial Machine health monitoring and Predictive maintenance solution leveraging Embedded system, Industrial IoT and Machine Learning Technologies by finding Remaining useful life time of various Machines used in production process.



2.2 About upskill Campus (USC)

2.3 The IoT Academy

The IoT academy is EdTech Division of UCT that is running long executive certification programs in collaboration with EICT Academy, IITK, IITR and IITG in multiple domains.

2.4 Objectives of this Internship program

The objective for this internship program was to

☛ get practical experience of working in the industry.

☛ to solve real world problems.

☛ to have improved job prospects.

☛ to have Improved understanding of our field and its applications.

☛ to have Personal growth like better communication and problem solving.

2.5 Reference

[1] Géron, A. *Hands-On Machine Learning with Scikit-Learn, Keras & TensorFlow*, 3rd Edition, O'Reilly Media, 2022.

[2] Pedregosa, F., et al. *Scikit-learn: Machine Learning in Python*. Journal of Machine Learning Research (JMLR), 2011.

[3] Flask Documentation. https://flask.palletsprojects.com/

2.6 Glossary

Terms	Acronym
Artificial Intelligence	AI
Machine Learning	ML
Credit Card Fraud Detection	CCFD
Receiver Operating Characteristic	ROC
Application Programming Interface	API

3 Problem Statement

Credit card fraud has become a significant challenge for banks and financial institutions due to the rapid growth of online transactions and digital payment systems. Fraudulent transactions result in financial losses, reduced customer trust, and increased operational costs. Traditional rule-based fraud detection methods often fail to identify new and evolving fraud patterns, leading to false positives and missed fraudulent activities.

The objective of this project is to develop an **AI-powered Credit Card Fraud Detection Engine** that can accurately identify fraudulent transactions using **Machine Learning** techniques. The system analyzes transaction details such as **transaction amount, transaction time, and device/location deviation score** to classify transactions as either legitimate or fraudulent. The proposed solution aims to improve fraud detection accuracy, minimize financial losses, and provide a secure and efficient transaction monitoring system.

4 Existing and Proposed solution

4.1.1 Existing Solution

Traditional fraud detection systems use **rule-based methods** to identify suspicious transactions. These systems require frequent updates, generate false positives, and struggle to detect new fraud patterns.

4.1.2 Proposed Solution

The proposed **Credit Card Fraud Detection AI Engine** uses **Machine Learning** to detect fraudulent transactions based on transaction amount, time, and device/location behavior. It is developed using **Python, Flask, Scikit-learn, Logistic Regression, and StandardScaler** for accurate real-time fraud prediction.

4.1.3 Value Addition

- Improved fraud detection accuracy.
- Reduced false positives.
- Real-time fraud prediction.
- Interactive dashboard for easy monitoring.

4.2 Code submission (Github link)

https://github.com/nandhuanandhu047-collab/upskillcampus/blob/main/app.py

4.3 Report submission (Github link)

https://github.com/nandhuanandhu047-collab/upskillcampus/blob/main/CreditCardFraudDetectionAIEngine_Anandhu_USC_UCT.pdf

5 Proposed Design/ Model

The proposed system is a machine learning-based web application for detecting fraudulent credit card transactions. The application accepts transaction details, preprocesses the data using StandardScaler, predicts fraud using a Logistic Regression model, and displays the result through a Flask-based dashboard.

High Level Diagram (if applicable)

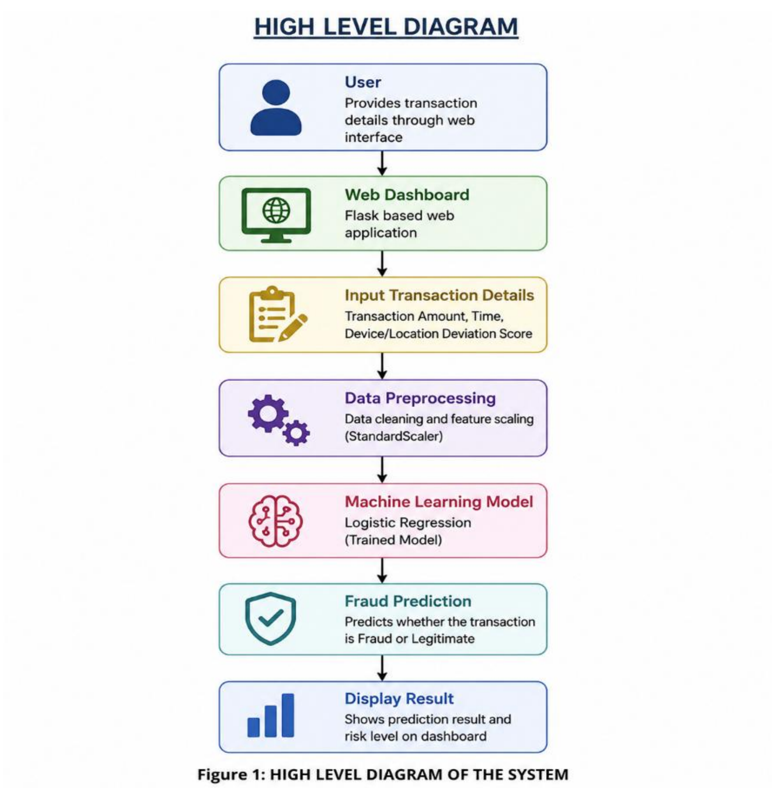


Figure 1: HIGH LEVEL DIAGRAM OF THE SYSTEM

Figure 1: HIGH LEVEL DIAGRAM OF THE SYSTEM

5.1

Interfaces

The system provides a simple web-based interface developed using **Flask** for entering transaction details. User inputs are securely processed and forwarded to the machine learning model for fraud prediction. The prediction results are displayed instantly through an interactive dashboard with clear risk classification. This interface ensures fast, accurate, and user-friendly fraud detection.

6 Performance Test

The performance of the **Credit Card Fraud Detection AI Engine** was evaluated based on prediction accuracy, response time, and reliability. The system was tested using different transaction scenarios to verify its ability to identify fraudulent transactions efficiently.

Identified Constraints

- Imbalanced dataset with fewer fraud cases.
- Prediction accuracy and response time.

Handling the Constraints

- Applied **StandardScaler** for feature normalization.
- Used **Logistic Regression** for fast and efficient prediction.

Performance Results

- Fast real-time prediction through the Flask application.
- Good fraud detection accuracy with low response time.

6.1 Test Plan

The system was tested using various transaction inputs with different amounts, transaction times, and device/location deviation scores. Test cases included both legitimate and fraudulent transaction scenarios to verify prediction accuracy and system reliability.

6.2 Test Procedure

Transaction details were entered through the Flask web interface. The input data was preprocessed using **StandardScaler** and passed to the **Logistic Regression** model. The predicted result was then displayed on the dashboard as either **Legitimate** or **Fraudulent**.

6.3 Performance Outcome

The model successfully identified fraudulent transactions with good prediction accuracy and fast response time. The system demonstrated reliable performance and provided real-time fraud detection through an interactive dashboard, making it suitable for practical financial security applications.

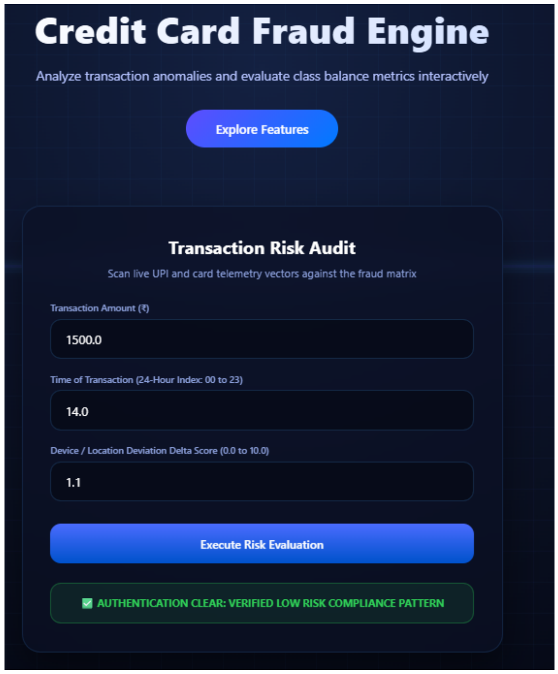


Fig 2 Legitimate Transaction Detection

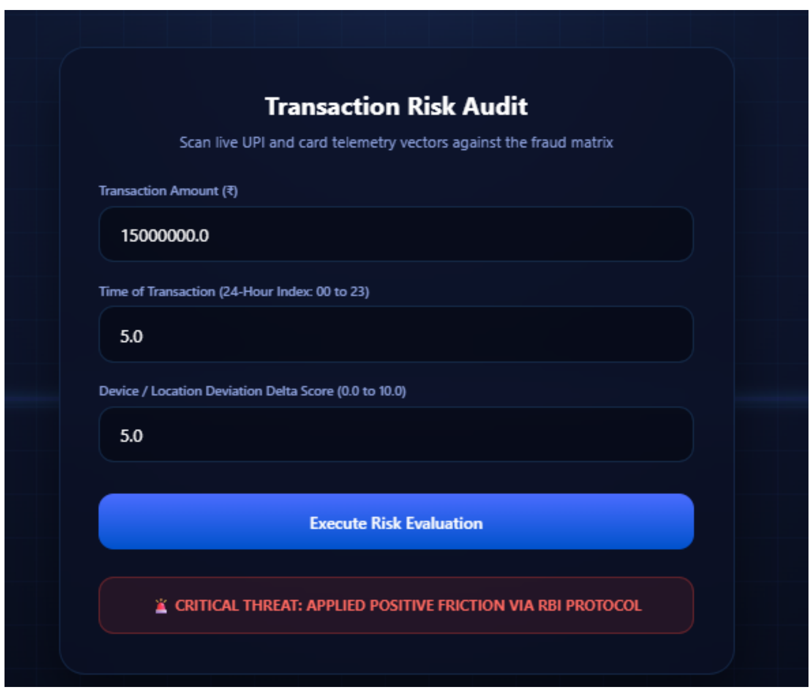


Fig 3 Fraudulent Transaction Detection

7 My learnings

During this internship, I gained practical knowledge of **Data Science, Machine Learning, Python, Flask, and Scikit-learn** by developing the **Credit Card Fraud Detection AI Engine**. I learned how to preprocess data, train machine learning models, evaluate their performance, and build a user-friendly web application. This experience improved my analytical thinking, problem-solving abilities, and project development skills, preparing me for a career in Data Science and Artificial Intelligence.

8 Future work scope

The **Credit Card Fraud Detection AI Engine** can be further enhanced by integrating advanced machine learning algorithms such as **Random Forest, XGBoost, and Deep Learning** to improve prediction accuracy. The system can also be connected to **real-time banking APIs** for live transaction monitoring and instant fraud detection. Additional features such as **user authentication, cloud deployment, explainable AI (XAI), and continuous model retraining** can be implemented to improve scalability, security, and overall performance in real-world financial applications.